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Analyzing Hiring Decisions and Fairness in Recruitment

INTRODUCTION

- This project analyzes recruitment data to understand hiring decisions
- The main **outcome** is whether a candidate is Selected or Rejected
- The goal is to identify key factors influencing hiring and evaluate fairness

PROBLEM & OBJECTIVE

- Hiring decisions are often complex and unclear
- Companies need data-driven insights to improve recruitment
- This project aims to:
  - Identify what drives hiring decisions
  - Detect potential bias or unfair patterns
  - Improve decision-making process

Dataset Overview

- APPLICANT_ID → UNIQUE IDENTIFIER FOR EACH CANDIDATE
- APPLICANT_NAME → CANDIDATE NAME
- GENDER → CANDIDATE GENDER
- UNIVERSITY → CANDIDATE'S UNIVERSITY
- EDUCATION_LEVEL → DEGREE LEVEL (E.G., BACHELOR, MASTER)
- CGPA → ACADEMIC PERFORMANCE SCORE
- EXPERIENCE_YEARS → YEARS OF WORK EXPERIENCE
- JOB_APPLIED → POSITION APPLIED FOR
- SKILLS_SCORE → TECHNICAL SKILL ASSESSMENT SCORE
- INTERVIEW_SCORE → INTERVIEW PERFORMANCE
- FINAL_DECISION → TARGET VARIABLE (SELECTED / REJECTED)

METHODOLOGY

- COLLECTING DATA
- CLEANING DATA
- EXPLORATORY DATA ANALYSIS (EDA)
- ANALYSING
- BUILD PREDICTIVE MODELING
- VISUALIZATION

TOOLS USED

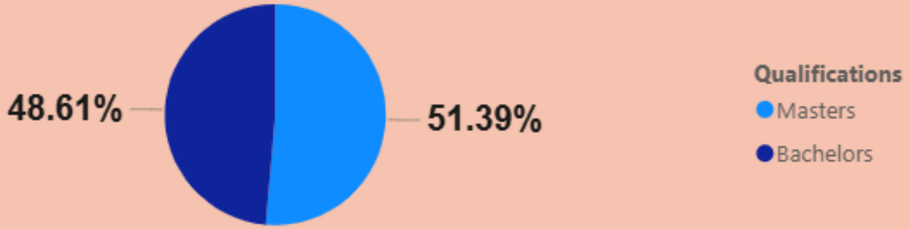
- PYTHON: CLEANING DATA, ANALYSIS, AND PREDICTION
- POWER BI: VISUALIZATION

2000
Total Candidates

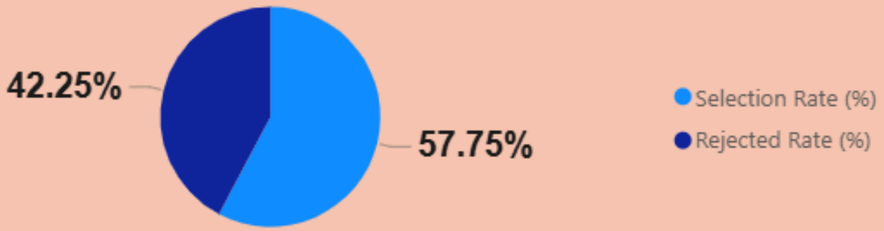
1155
Total Selected Candidates

57.75
Selection Rate (%)

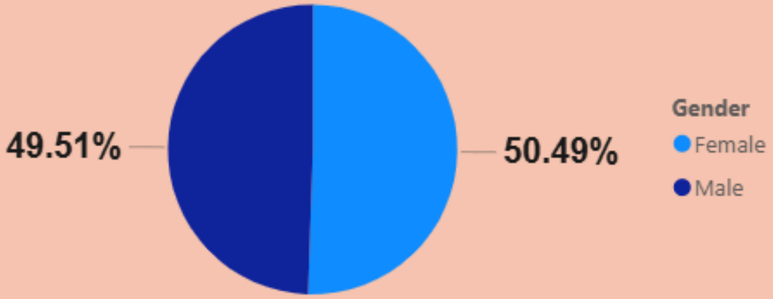
Selection Rate (%) by Qualifications



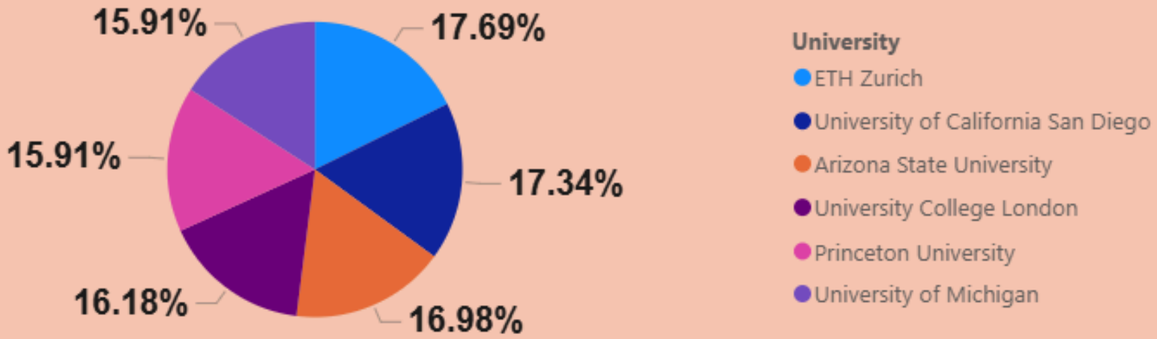
Selection Rate (%) and Rejected Rate (%)

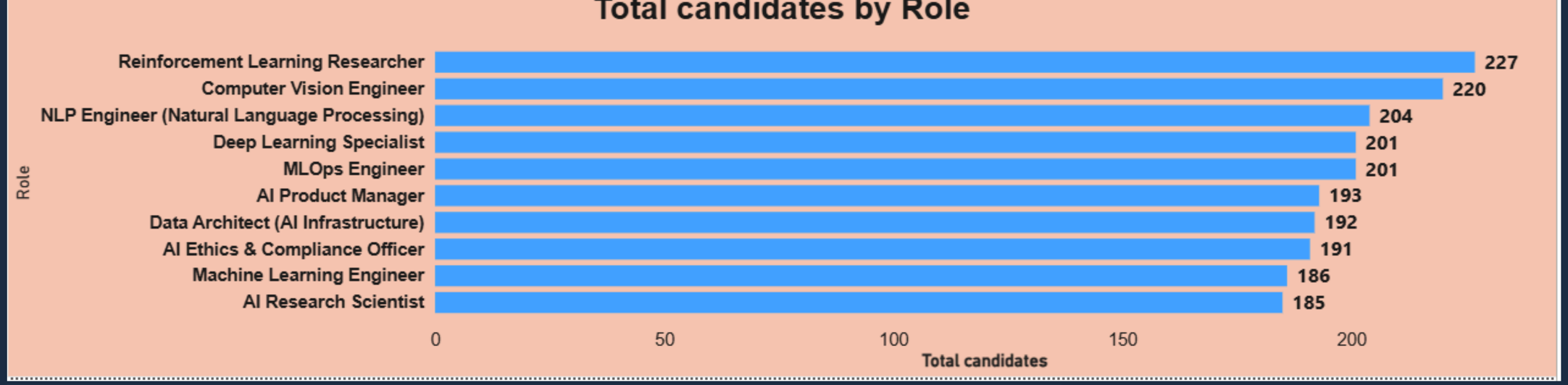


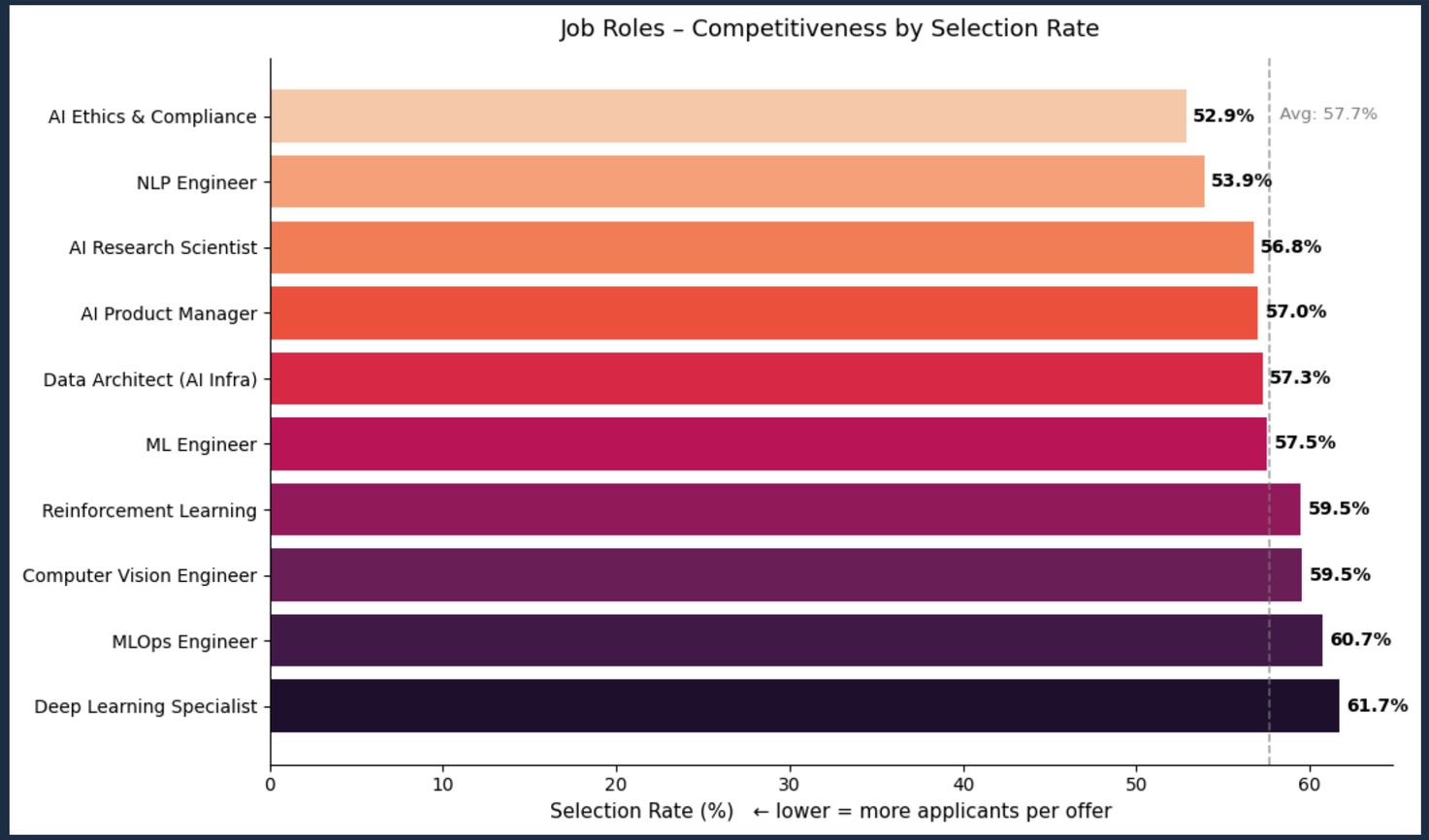
Selection Rate (%) by Gender



Selection Rate (%) by University







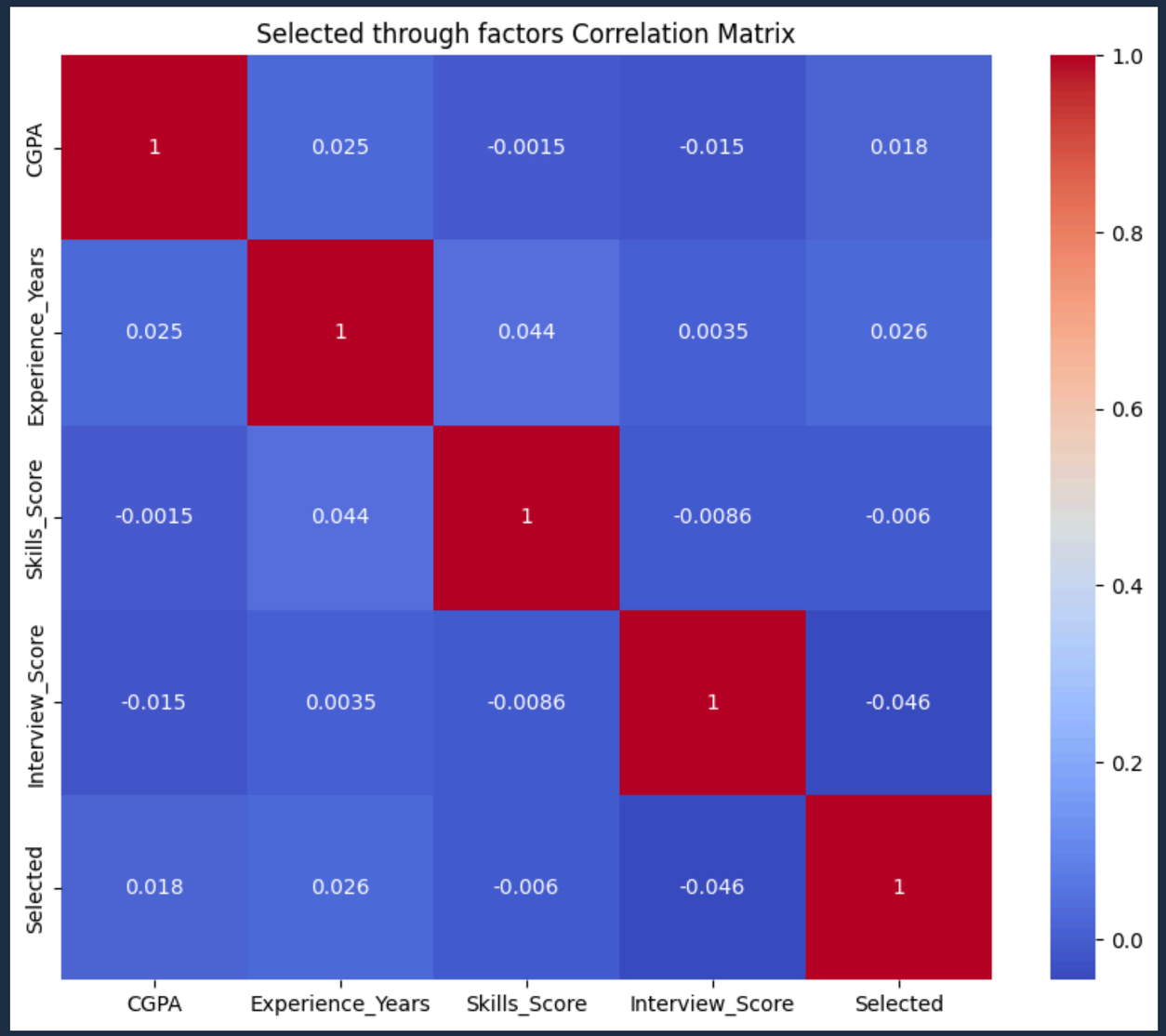
- FEW COMPETITIVE:

DEEP LEARNING SPECIALIST: 61.7%
MLOPS ENGINEER: 60.7%
COMPUTER VISION / REINFORCEMENT LEARNING: ~59.5%

- MOST COMPETITIVE:

AI ETHICS & COMPLIANCE: 52.9%
NLP ENGINEER: 53.9%
AI RESEARCH SCIENTIST: 56.8%

1. What drives hiring decisions



- No strong correlation between scores and selection
- Skills and Interview Scores show minimal differences

→ **Hiring decisions are not driven by a single factor**

What Really Predicts Getting Hired?
(Logistic Regression Coefficients)
Skills_Score
-0.000883
Interview_Score
-0.005959
Experience_Years
+0.048887
CGPA
+0.071738
0.00
0.01
0.02
0.03
0.04
0.05
0.06
0.07
Absolute Coefficient Magnitude
→ CGPA & Experience_Years are now the dominant positive drivers.

Q&A TIME

Thank You

for your time and attention